

Supporting Information for "Climate-dependency of Impact of Increased Carbon Dioxide on African Monsoon Rainfall: Insights from model simulations"

Xiaoxu Shi* (1, 2), Martin Werner (2), Francesco S.R. Pausata (3), Jiping Liu* (4, 1), Roberta D'Agostino (5), Yongyun Hu (6), Hu Yang (1), Hang Wang (4, 1), Chaoyuan Yang (1), Gerrit Lohmann (2)

¹Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai, China

²Alfred Wegener Institute, Helmholtz Center for Polar and Marine Research, Klussmanstr. 3, 27570

Bremerhaven, Germany

³University of Quebec in Montreal, Montreal (QC), Canada

⁴School of Atmospheric Sciences, Sun Yat-sen University, Zhuhai, China

⁵Italian National Research Council, Institute of Atmospheric Sciences and Climate, via per Monteroni Km

1.2, 73100, Lecce, Italy

⁶Peking University, Peking, China

Contents of this file

1. Text S1 to S2
2. Figures S1 to S12

Introduction

The texts and figures shown here provide the supporting information for the main manuscript.

Text S1 describes the experimental design.

Text S2 describes the moisture budget analysis.

Fig S1: Spatial resolution of the ocean model FESOM.

Fig S2: Surface air temperature anomalies relative to PI in other reference simulations.

Fig S3: Southern Hemisphere sea ice concentration in reference simulations.

Fig S4: Simulated PI precipitation.

Fig S5: Simulated precipitation anomalies between paleo and PI versus reconstruction.

Fig S6: Surface air temperature anomalies during JJAS between sensitivity and reference simulations.

Fig S7: Spatial patterns of thermodynamic and dynamic components and net precipitation changes.

Fig S8: Simulated anomalies of zonal-mean meridional stream function (average over all longitudes) between sensitivity and reference simulations.

Fig S9: Lower atmospheric thickness calculated as the difference between the 700 and 925 hPa geopotential heights for all reference simulations.

Fig S10: AMOC streamfunction and AMOC strength for all simulations.

Fig S11: Composite analysis for ASM precipitation anomalies versus AMOC indices.

Corresponding author: Xiaoxu Shi (shixiaoxu@sml-zhuhai.cn) and Jiping Liu (li-ujp63@mail.sysu.edu.cn),

Fig S12: Simulated mean ASM precipitation against the NARI (difference in surface surface temperature between the subtropical North Atlantic and the global tropical oceans).

Text S1 Experimental design We first perform 5 equilibrium simulations, representing 3 interglacial time periods, i.e., pre-industrial (PI), mid-Holocene (MH), and last interglacial (LIG), as well as 2 glacial periods, i.e., Last Glacial Maximum (LGM) and Marine Isotope Stage 3 (MIS3). The initial conditions for the atmosphere in our PI simulation are derived from the Atmospheric Model Intercomparison Project (AMIP) (Roeckner et al., 2004). The ocean model is initialized with the World Ocean Atlas (WOA) climatological temperature and salinity data for the years 1950-2000 (Levitus et al., 2010). We run the PI simulation for 1,500 model years with dynamic vegetation. The MH and LIG simulations are initialized from the PI run. The boundary conditions are configured following the criteria of PMIP4 (Otto-Bliesner et al., 2017). Orbital parameters are calculated following Berger (1977), and the greenhouse gas concentrations are taken from multi-archive reconstructions from ice core records and recent measurements of firn air and atmospheric samples (Flückiger et al., 2002; Monnin et al., 2004; Schilt et al., 2010; Buiron et al., 2011; Schneider et al., 2013; Köhler et al., 2017). The CO₂ concentration used in our study is 284.32 ppm for PI, 264.4 ppm for MH, 275 ppm for LIG, 210.5 ppm for MIS3, and 190 ppm for LGM. For LGM and MIS3, we fix the boundary conditions at 21 ka and 38 ka respectively. The topography and ice-sheet properties are derived from the GLAC1D reconstruction (Tarasov & Peltier, 2003; Tarasov et al., 2012; Briggs et al., 2014). Both LGM and MIS3 experiments are initialized from a previous LGM model study (Werner et al., 2016). All of the 4 paleo simulations are integrated for 1,000 model years, with the simulated climate being in a quasi-equilibrium state for the final 100 model years. Then, for each experiment, we double the CO₂ concentration, while keeping other boundary conditions unchanged, and further integrate for 200 model years. We observe no obvious trends in surface air temperature and sea ice in the last 50 model years. The experiments configured with rising CO₂ are labeled as PI_CO₂, MH_CO₂, LIG_CO₂, LGM_CO₂, and MIS3_CO₂, respectively.

Text S2 Moisture budget analysis The linearized anomalous moisture budget is decomposed into thermodynamic, dynamic terms and a residual (Res) component as:

$$\rho_w g \delta(P - E) = - \int_0^{p_s} \Delta \cdot (\delta \bar{q} \bar{\mathbf{u}}_{ref}) dp - \int_0^{p_s} \Delta \cdot (\bar{q}_{ref} \delta \bar{\mathbf{u}}) dp - Res \quad (1)$$

where P and E denote precipitation and evaporation, respectively, ρ_w is the water density, p pressure, q specific humidity, $\bar{\mathbf{u}}$ the horizontal vector wind, and Res represents residual term that includes contributions from transient eddies and surface processes linked to topography. $\bar{\mathbf{u}}_{ref}$ and $\bar{q}_{ref}$, respectively, represents the mean horizontal vector wind and the mean specific humidity over the production period of each reference simulation. δ indicates the difference between each sensitivity experiment (with rising CO₂) and the corresponding reference climate. For example, $\delta \bar{\mathbf{u}}$ ($\delta \bar{q}$) refers to the difference in mean horizontal wind (specific humidity) between sensitivity and reference simulations. In equation (1), the first term on the right-hand side, $-\int_0^{p_s} \Delta \cdot (\delta \bar{q} \bar{\mathbf{u}}_{ref}) dp$, is the thermodynamic contribution (dTH) related to changes in moisture flux convergence, while the second term, $-\int_0^{p_s} \Delta \cdot (\bar{q}_{ref} \delta \bar{\mathbf{u}}) dp$, accounts for the dynamic contribution (dDY), representing changes in winds with unchanged moisture and is mostly associated with changes in the mean atmospheric flow.

References

- Berger, A. (1977). Long-term variations of the earth's orbital elements. *Celestial Mechanics*, 15(1), 53–74.
- Briggs, R. D., Pollard, D., & Tarasov, L. (2014). A data-constrained large ensemble analysis of antarctic evolution since the eemian. *Quaternary Science Reviews*, 103, 91–115.
- Buiron, R. D., Chappellaz, J., Stenni, B., Frezzotti, M., Baumgartner, M., Capron, E., ... others (2011). Taldice-1 age scale of the talos dome deep ice core, east antarctica. *Climate of the Past*, 7(1), 1–16.

- Flückiger, J., Monnin, E., Stauffer, B., Schwander, J., Stocker, T. F., Chappellaz, J., ... Barnola, J.-M. (2002). High-resolution holocene n₂o ice core record and its relationship with ch₄ and co₂. *Global Biogeochemical Cycles*, 16(1), 10–1.
- Köhler, P., Nehrbass-Ahles, C., Schmitt, J., Stocker, T. F., & Fischer, H. (2017). A 156 kyr smoothed history of the atmospheric greenhouse gases co₂, ch₄, and n₂o and their radiative forcing. *Earth System Science Data*, 9(1), 363–387.
- Levitus, S., Locarnini, R. A., Boyer, T. P., Mishonov, A. V., Antonov, J. I., Garcia, H. E., ... Seidov, D. (2010). World ocean atlas 2009.
- Monnin, E., Steig, E. J., Siegenthaler, U., Kawamura, K., Schwander, J., Stauffer, B., ... others (2004). Evidence for substantial accumulation rate variability in antarctica during the holocene, through synchronization of co₂ in the taylor dome, dome c and dml ice cores. *Earth and Planetary Science Letters*, 224(1-2), 45–54.
- Otto-Bliesner, B., Braconnot, P., Harrison, S., Lunt, D., Abe-Ouchi, A., Albani, S., ... others (2017). The pmip4 contribution to cmip6–part 2: Two interglacials, scientific objective and experimental design for holocene and last interglacial simulations. *Geoscientific Model Development*, 10, 3979–4003.
- Roeckner, E., Brokopf, R., Esch, M., Giorgetta, M., Hagemann, S., Kornblueh, L., ... Schulzweida, U. (2004). The atmospheric general circulation model echam5 part ii: Sensitivity of simulated climate to horizontal and vertical resolution. *Report / Max-Planck-Institut für Meteorologie*, 354.
- Schilt, A., Baumgartner, M., Blunier, T., Schwander, J., Spahni, R., Fischer, H., & Stocker, T. F. (2010). Glacial–interglacial and millennial-scale variations in the atmospheric nitrous oxide concentration during the last 800,000 years. *Quaternary Science Reviews*, 29(1-2), 182–192.
- Schneider, R., Schmitt, J., Köhler, P., Joos, F., & Fischer, H. (2013). A reconstruction of atmospheric carbon dioxide and its stable carbon isotopic composition from the penultimate glacial maximum to the last glacial inception. *Climate of the Past*, 9(6), 2507–2523.
- Tarasov, L., Dyke, A. S., Neal, R. M., & Peltier, W. R. (2012). A data-calibrated distribution of deglacial chronologies for the north american ice complex from glaciological modeling. *Earth and Planetary Science Letters*, 315, 30–40.
- Tarasov, L., & Peltier, W. R. (2003). Greenland glacial history, borehole constraints, and eemian extent. *Journal of Geophysical Research: Solid Earth*, 108(B3).
- Werner, M., Haese, B., Xu, X., Zhang, X., Butzin, M., & Lohmann, G. (2016). Glacial–interglacial changes in h₂18o, hdo and deuterium excess—results from the fully coupled echam5/mpi-om earth system model. *Geoscientific Model Development*, 9(2), 647–670.

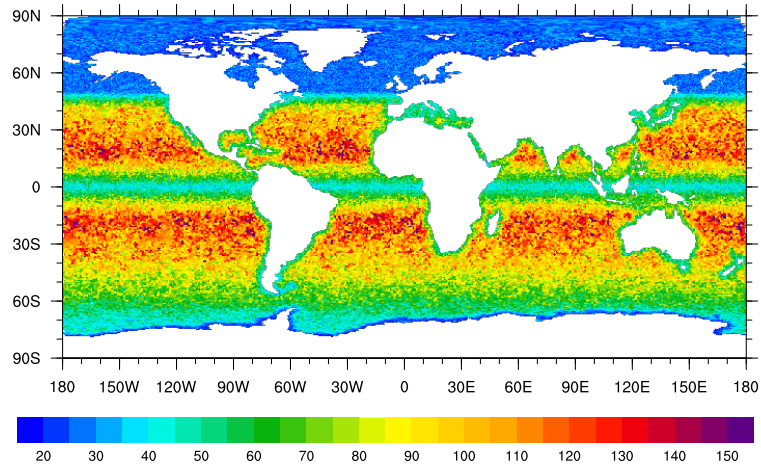


Figure S1. Spatial resolution of the ocean model FESOM. Units are km.

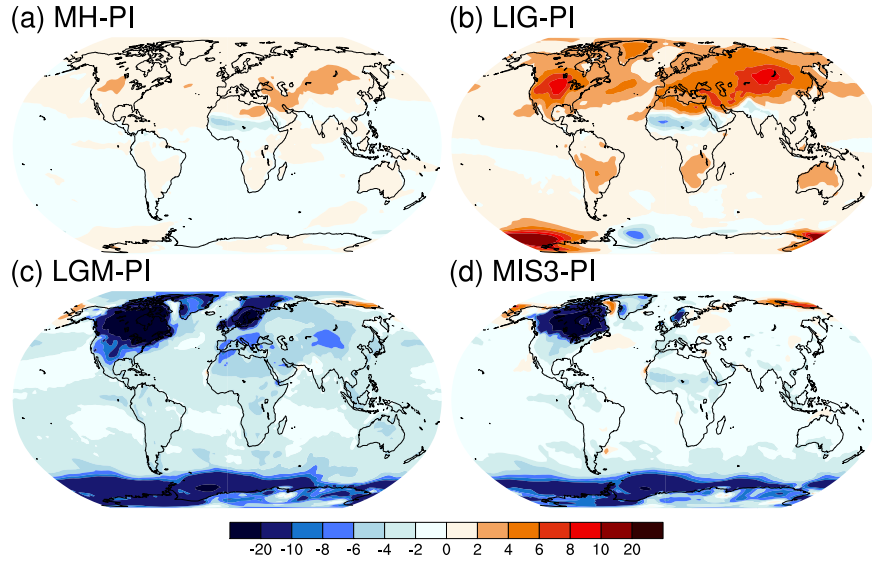


Figure S2. Surface air temperature anomalies relative to PI in (a) MH, (b) LIG, (c) LGM, and (d) MIS3. Units: °C.

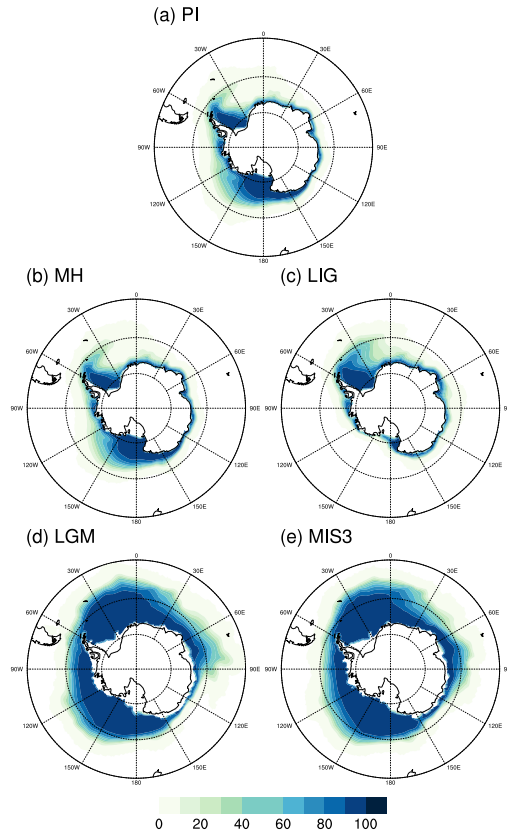


Figure S3. Southern Hemisphere sea ice concentration in (a) PI, (b) MH, (c) LIG, (d) LGM, and (e) MIS3. Units: %.

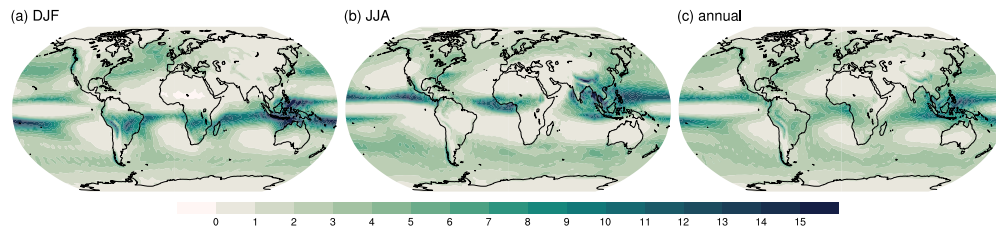


Figure S4. Simulated precipitation in PI for (a) DJF, (b) JJA, and (c) annual mean. Units: mm/day

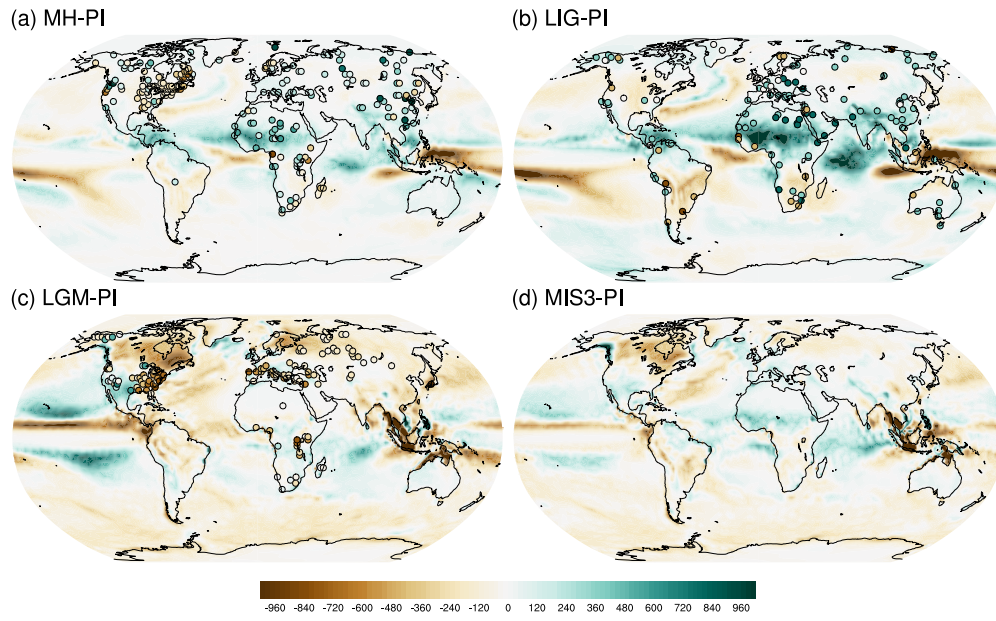


Figure S5. Simulated annual mean precipitation anomalies compared to PI in (a) MH, (b) LIG, (c) LGM, and (d) MIS3. Circles represent proxy-based reconstructed values. Units: mm/year

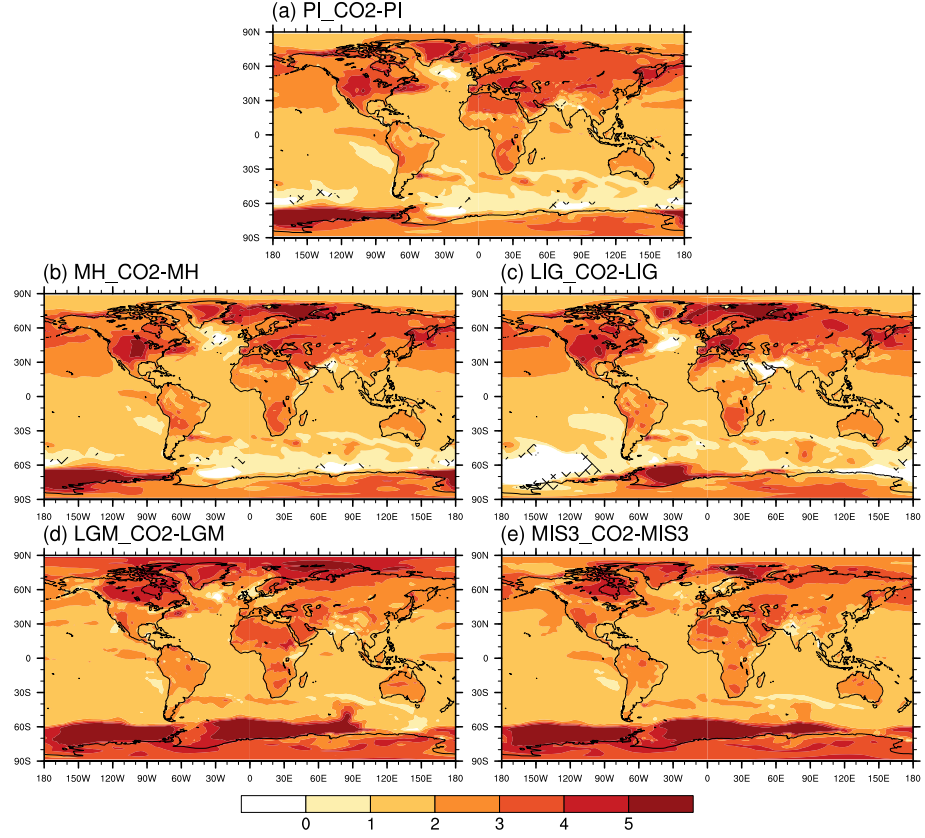


Figure S6. Surface air temperature anomalies (°C) during JJAS for (a) PL_CO₂ minus PI, (b) MH_CO₂ minus MH, (c) LIG_CO₂ minus LIG, (d) LGM_CO₂ minus LGM, and (e) MIS3_CO₂ minus MIS3. Non-slashed areas indicate significant changes at 95% confidence level based on Student's t test.

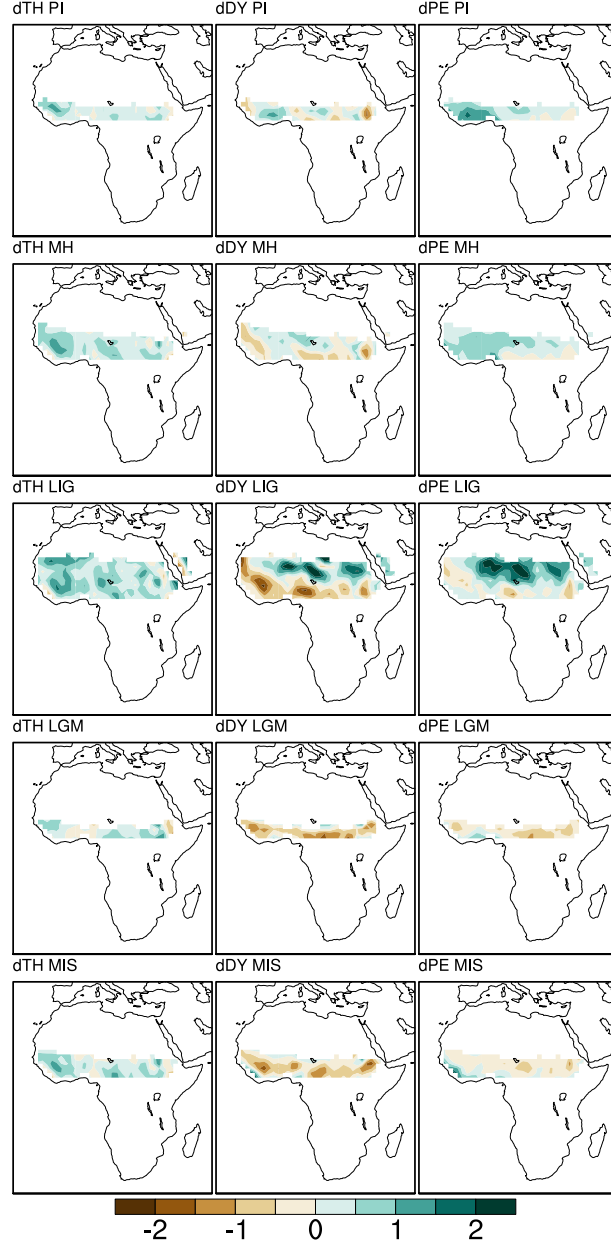


Figure S7. Spatial patterns of dTH (left panels), dDY (middle panels), and dPE (right panels) for each simulation. Only values for the monsoon domain (where JJAS-minus-DJFM precipitation anomaly exceeds 2 mm/day) are shown. Units: mm/day.

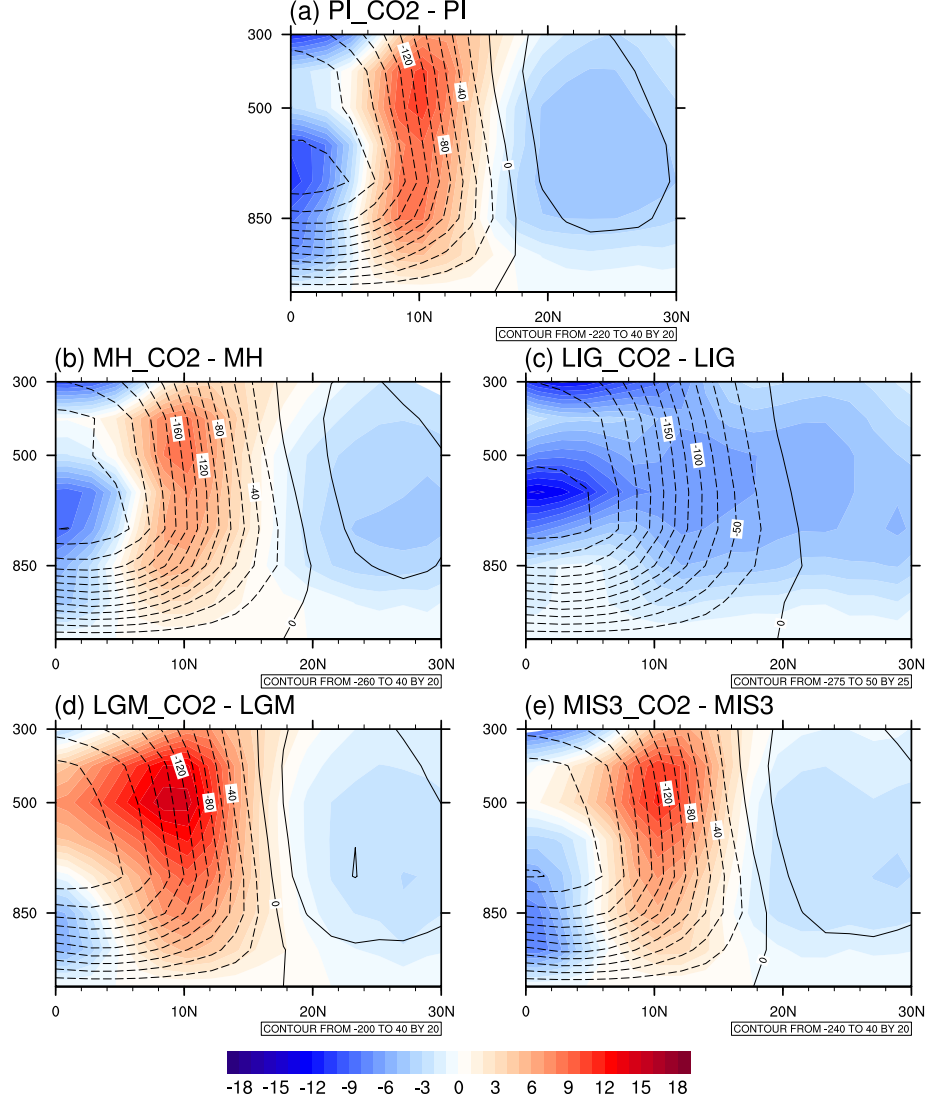


Figure S8. Simulated JJAS zonal-mean meridional stream function averaged over all longitudes (contours) and the anomalies of zonal-mean meridional stream function (shading) for (a) PI_CO_2 minus PI , (b) MH_CO_2 minus MH , (c) LIG_CO_2 minus LIG , (d) LGM_CO_2 minus LGM , and (e) MIS3_CO_2 minus MIS3 . Dashed/Solid lines stand for anticlockwise/clockwise circulation. Units: svp ($1 \text{ svp} = 10^9 \text{ kg/s}$).

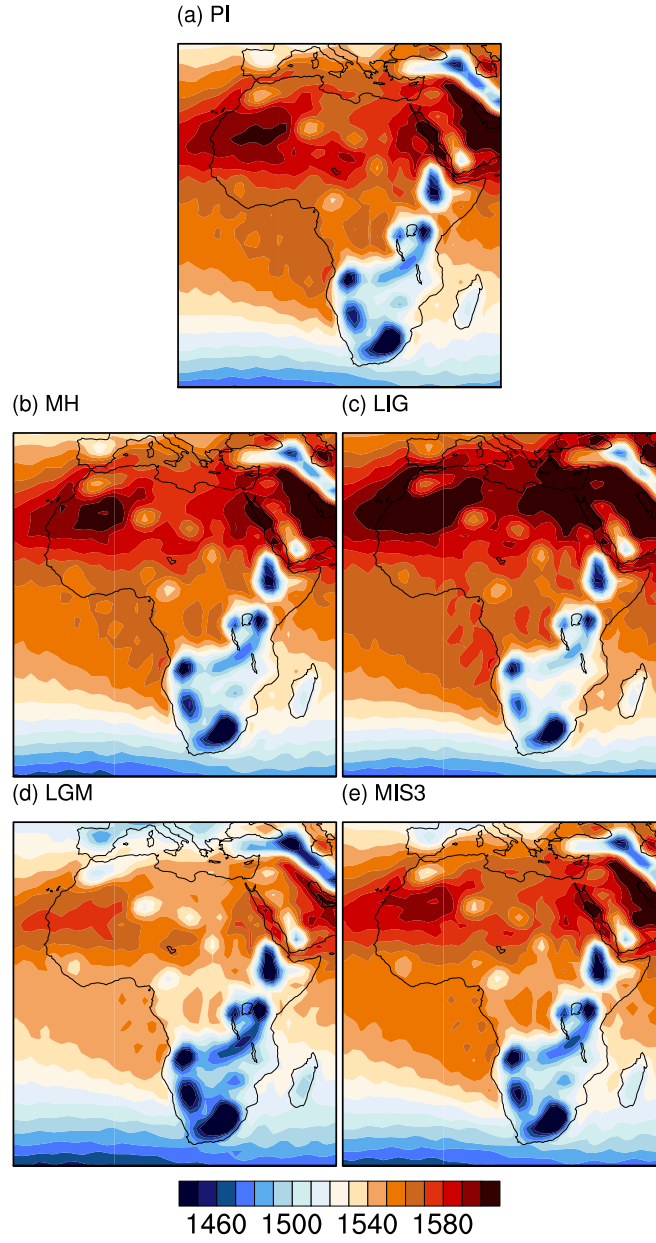


Figure S9. Lower atmospheric thickness in (a) PI, (b) MH, (c) LIG, (d) LGM, and (e) MIS3, calculated as the difference between the 700 and 925 hPa geopotential heights. Units: m.

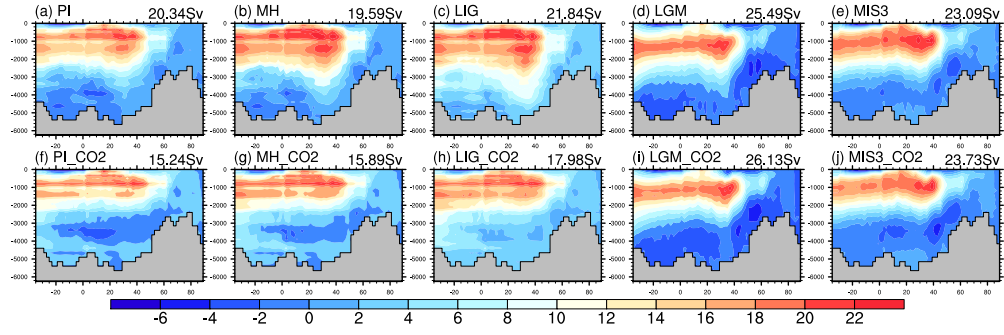


Figure S10. Simulated annual mean Atlantic meridional overturning circulation in all experiments. The top-right number in each panel represents the corresponding AMOC index. Units are Sv.

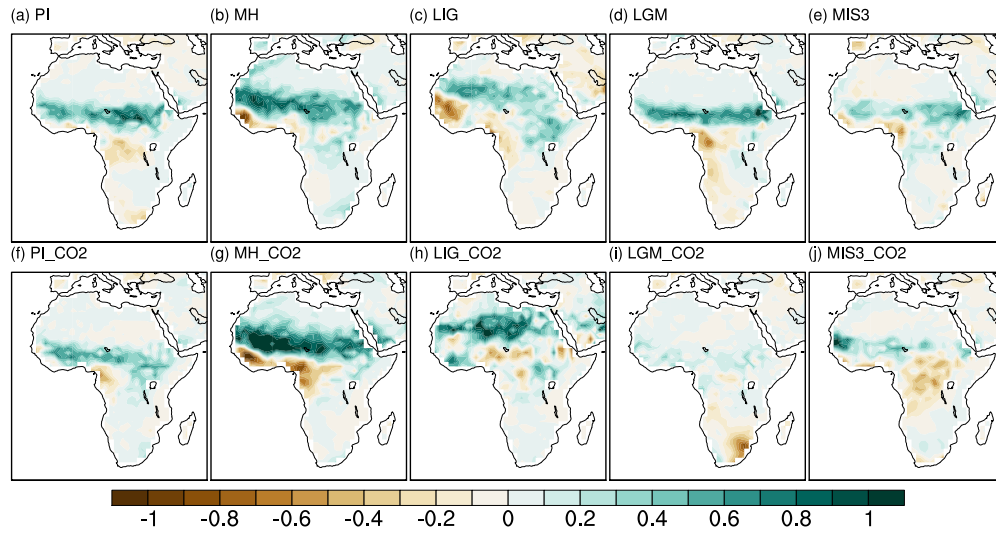


Figure S11. Simulated JJAS precipitation anomalies between high and low AMOC years (AMOC index being higher/lower than 1 standard deviation). Units: mm/day.

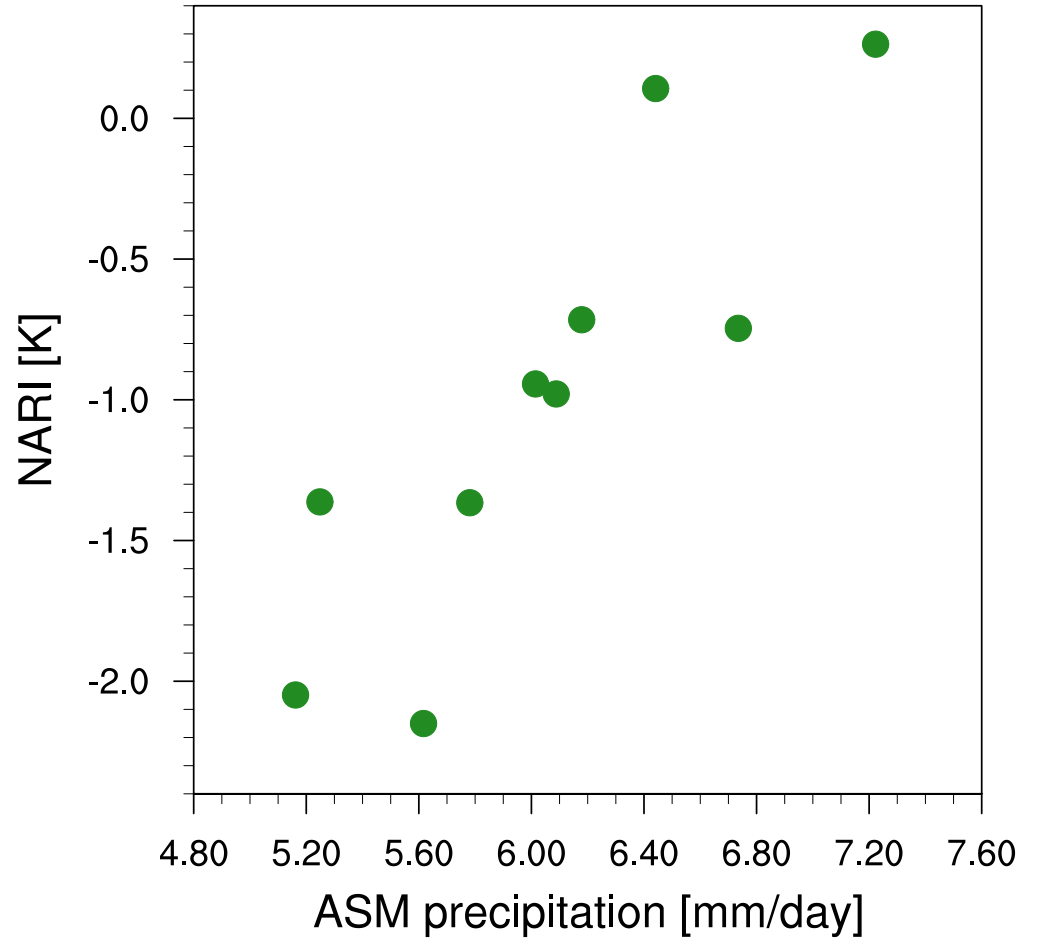


Figure S12. Simulated mean ASM precipitation in the 10 simulations in this study against the NARI (difference in surface surface temperature between the subtropical North Atlantic and the global tropical oceans). All averages are taken from July–September.